KAIZEN MATHS

Lesson starter

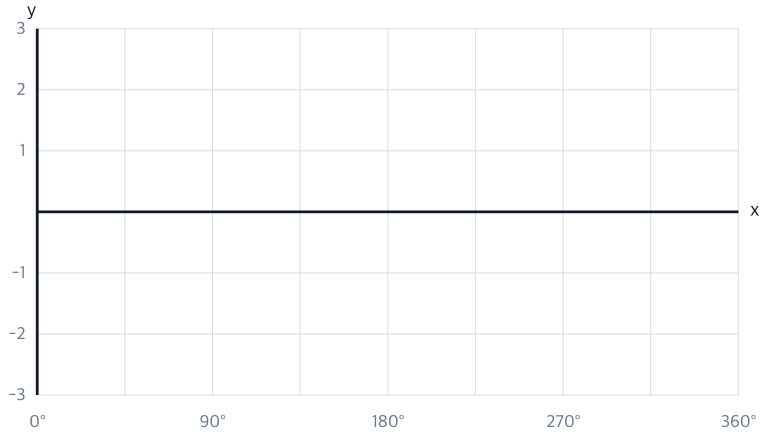
Trigonometric Graphs and Transformations

Single Transformations

Name _____

SA1

Sketch each graph and state the amplitude.



Sketch y = -2cos(x) and state the amplitude.

Show your working

KAIZEN MATHS

Lesson starter

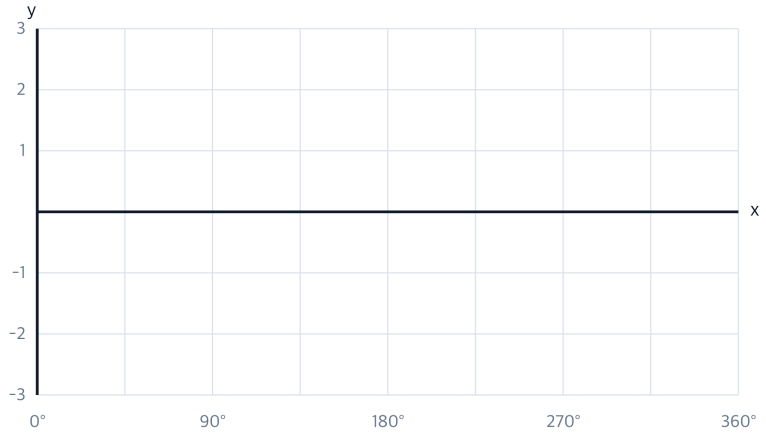
Trigonometric Graphs and Transformations

Single Transformations

Name _____

SB1

Sketch each graph and state the period.



Sketch y = cos(3(x)) and state the period.

Show your working

KAIZEN MATHS

Lesson starter

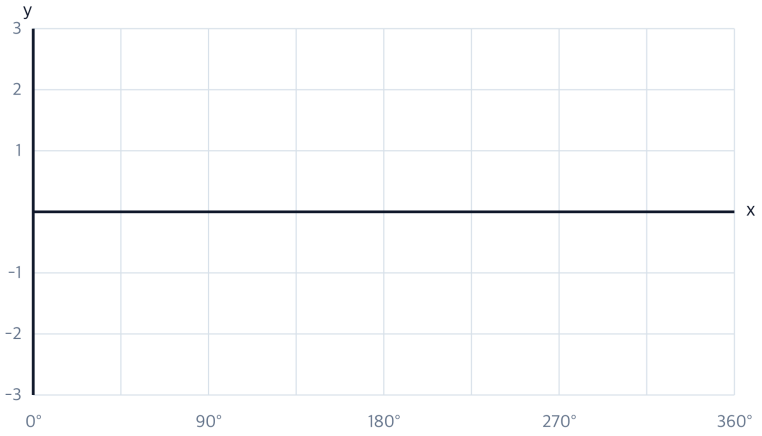
Trigonometric Graphs and Transformations

Single Transformations

Name _____

SC1

Sketch each graph and state the midline.



Sketch y = cos(x) + 1 and state the midline.

ALG-020E-L2 · SC

Show your working

KAIZEN MATHS

Lesson starter

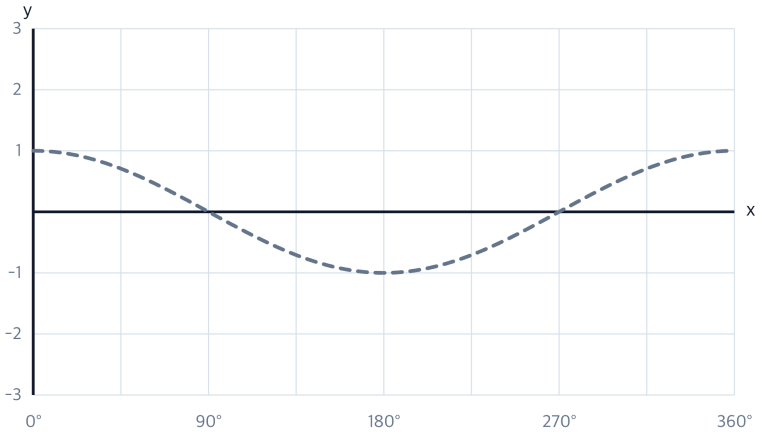
Trigonometric Graphs and Transformations

Single Transformations

Name _____

SD1

Sketch each reflected trigonometric graph.



parent graphimage to sketch

Sketch y = -cos(x) and describe the reflection.

ALG-020E-L2 · SD

Show your working

KAIZEN MATHS

Exit ticket

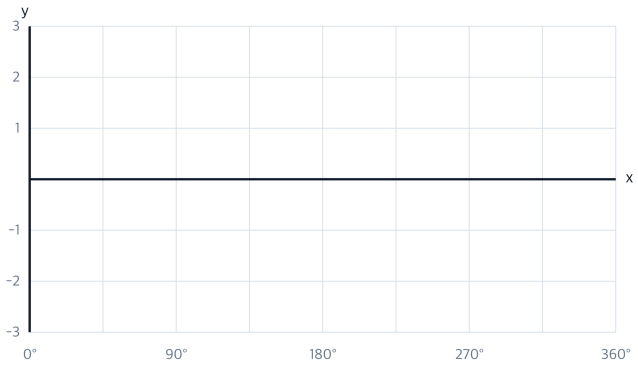
Trigonometric Graphs and Transformations

Single Transformations

Name _____

EA1

Sketch each graph and state the amplitude.



Sketch y = 2sin(x) and state the amplitude.

Show your working

KAIZEN MATHS

Exit ticket

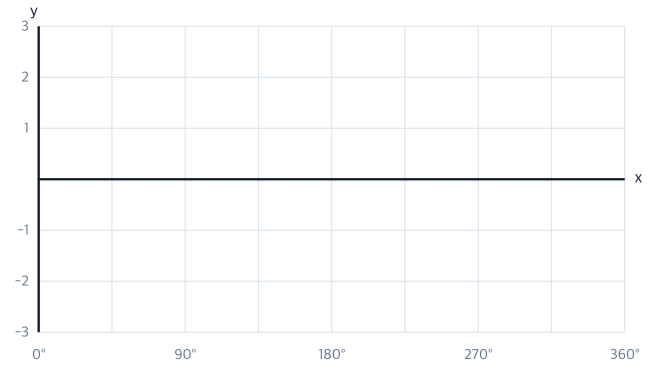
Trigonometric Graphs and Transformations

Single Transformations

Name _____

EB1

Sketch each graph and state the period.



Sketch y = cos(2(x)) and state the period.

Show your working

KAIZEN MATHS

Exit ticket

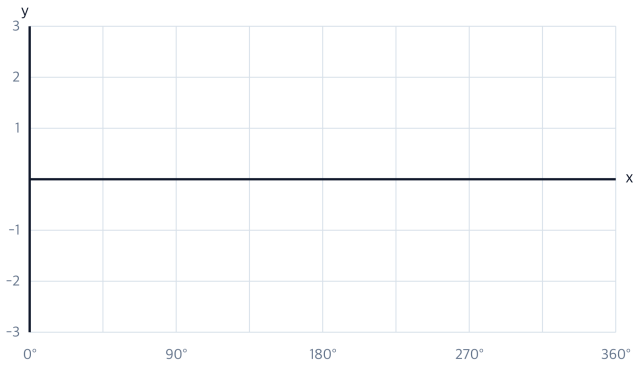
Trigonometric Graphs and Transformations

Single Transformations

Name _____

EC1

Sketch each graph and state the midline.



Sketch y = sin(x) - 1 and state the midline.

Show your working

KAIZEN MATHS

Exit ticket

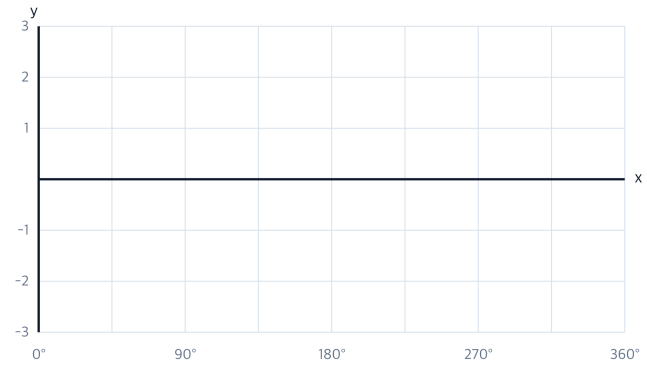
Trigonometric Graphs and Transformations

Single Transformations

Name _____

EA1

Sketch each graph and state the amplitude.



Sketch y = 2sin(x) and state the amplitude.

Show your working

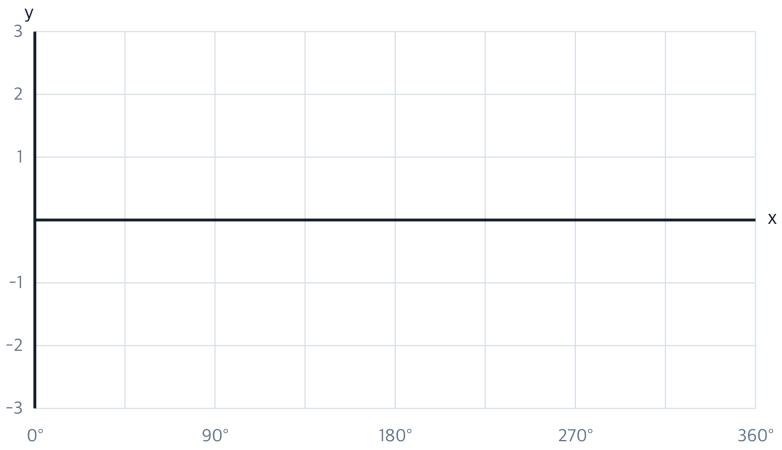
Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Sketch each graph and state the amplitude.



Sketch y = -2cos(x) and state the amplitude.

Answer:

Amplitude = 2. Period = 360°.

Working:

Start from the parent graph y = cos x.

The coefficient outside the trigonometric function is -2, so the amplitude is |-2| = 2.

The coefficient inside the bracket is 1, so the period is 360° ÷ 1 = 360°.

The graph is shifted 0° horizontally.

The + 0 outside the function gives the midline y = 0.

Summarise the graph features.

Amplitude	2
Period	360°
Midline	y = 0
Phase shift	none

Use the transformed key points to guide the sketch.

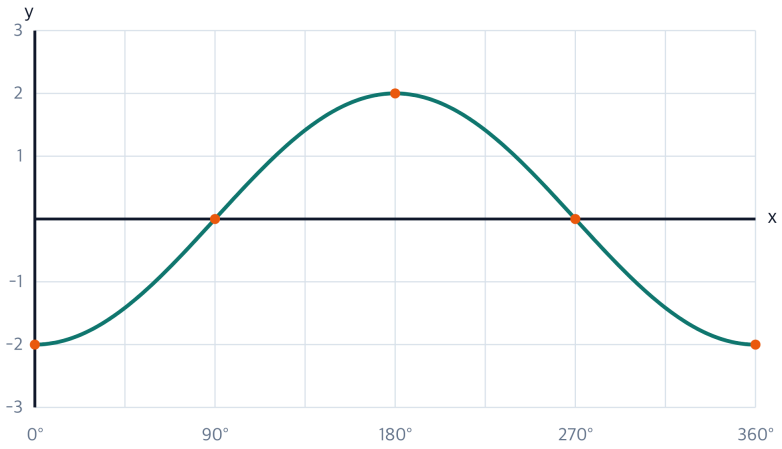
x	y
0°	-2
90°	0
180°	2
270°	0
360°	-2

Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1 (continued)



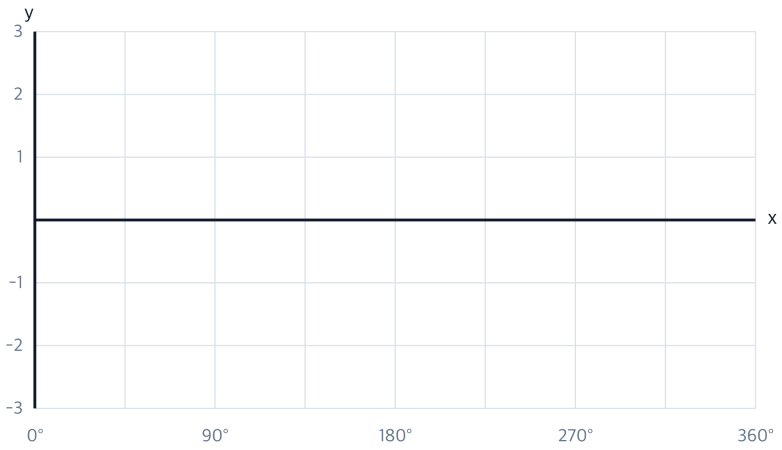
Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1

Sketch each graph and state the period.



Sketch y = cos(3(x)) and state the period.

Answer:

Period = 120°.

Working:

Start from the parent graph y = cos x.

The coefficient outside the trigonometric function is 1, so the amplitude is |1| = 1.

The coefficient inside the bracket is 3, so the period is 360° ÷ 3 = 120°.

The graph is shifted 0° horizontally.

The + 0 outside the function gives the midline y = 0.

Summarise the graph features.

Amplitude	1
Period	120°
Midline	y = 0
Phase shift	none

Use the transformed key points to guide the sketch.

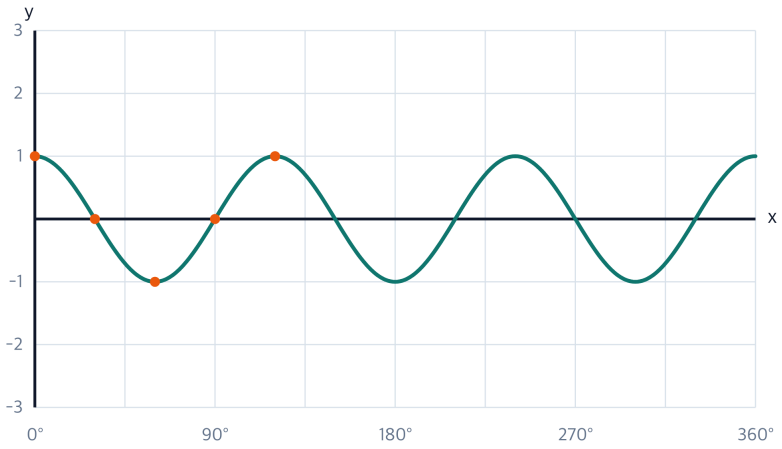
x	y
0°	1
30°	0
60°	-1
90°	0
120°	1

Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1 (continued)



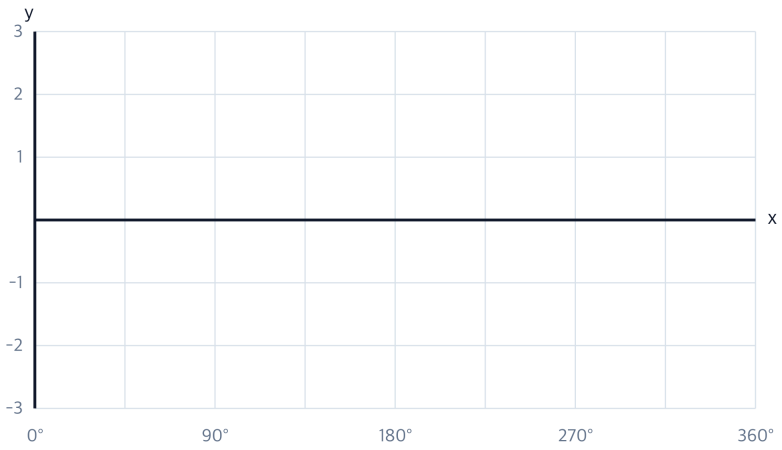
Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1

Sketch each graph and state the midline.



Sketch y = cos(x) + 1 and state the midline.

Answer:

Midline: y = 1.

Working:

Start from the parent graph y = cos x.

The coefficient outside the trigonometric function is 1, so the amplitude is |1| = 1.

The coefficient inside the bracket is 1, so the period is 360° ÷ 1 = 360°.

The graph is shifted 0° horizontally.

The + 1 outside the function gives the midline y = 1.

Summarise the graph features.

Amplitude	1
Period	360°
Midline	y = 1
Phase shift	none

Use the transformed key points to guide the sketch.

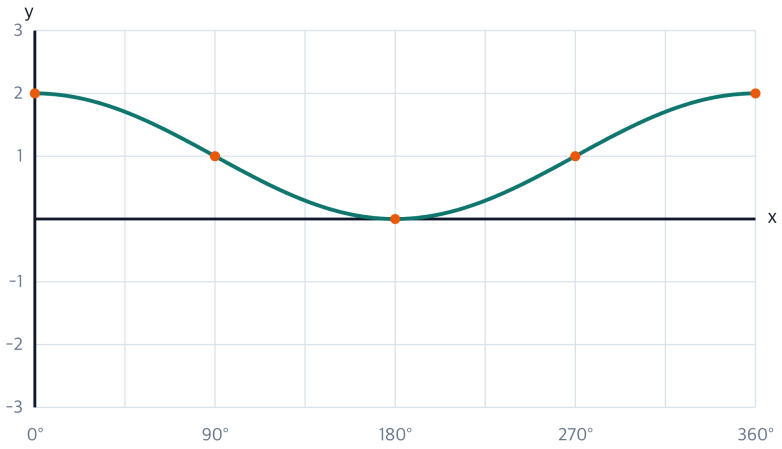
x	y
0°	2
90°	1
180°	0
270°	1
360°	2

Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1 (continued)



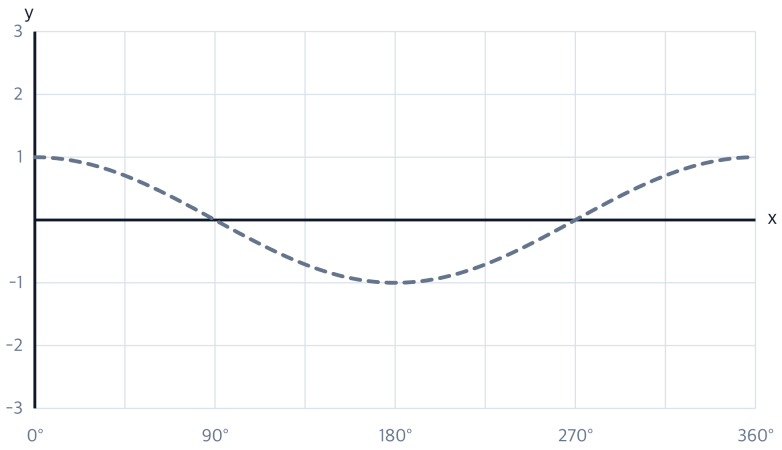
Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1

Sketch each reflected trigonometric graph.



parent graphimage to sketch

Sketch y = -cos(x) and describe the reflection.

Answer:

Reflection in the x-axis.

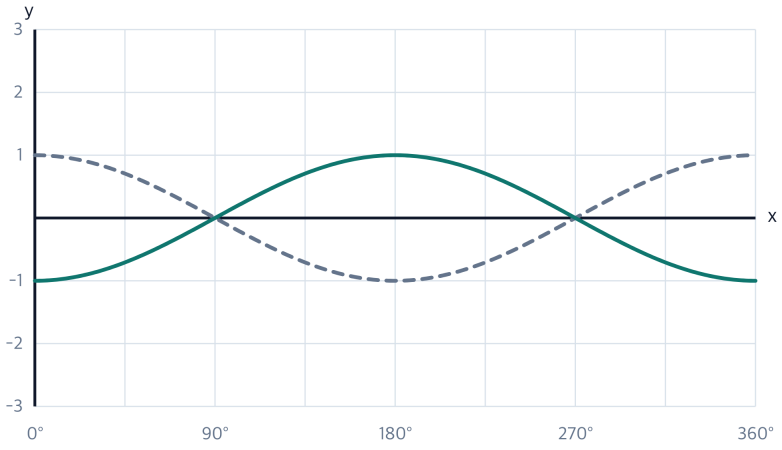
Working:

Start with y = cos x.

The negative sign outside the function multiplies every y-coordinate by -1.

This reflects the graph in the x-axis.

The period is unchanged at 360°.



parent graphy = -cos(x)

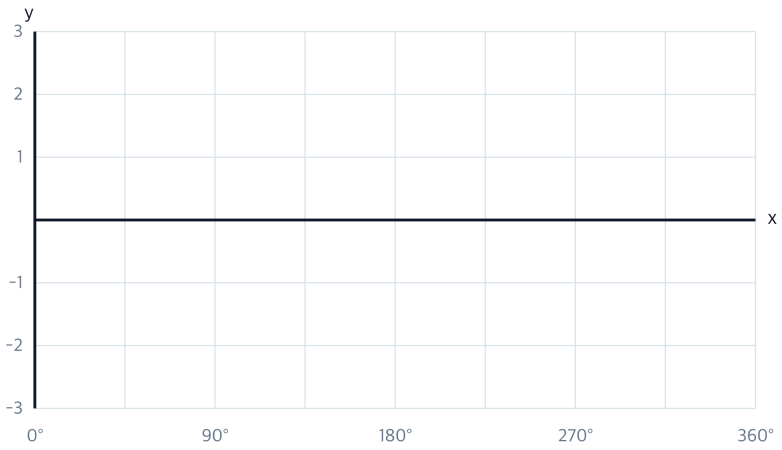
Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

Sketch each graph and state the amplitude.



Sketch y = 2sin(x) and state the amplitude.

Answer:

Amplitude = 2. Period = 360°.

Working:

Start from the parent graph y = sin x.

The coefficient outside the trigonometric function is 2, so the amplitude is |2| = 2.

The coefficient inside the bracket is 1, so the period is 360° ÷ 1 = 360°.

The graph is shifted 0° horizontally.

The + 0 outside the function gives the midline y = 0.

Summarise the graph features.

Amplitude	2
Period	360°
Midline	$y = 0$
Phase shift	none

Use the transformed key points to guide the sketch.

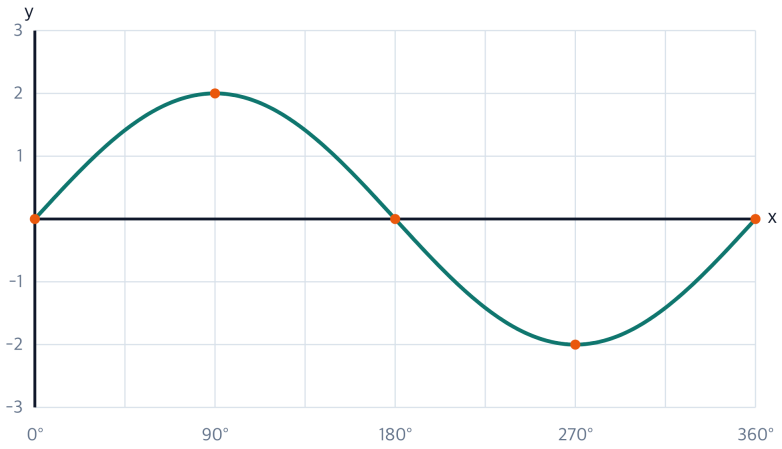
x	y
0°	0
90°	2
180°	0
270°	-2
360°	0

Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1 (continued)



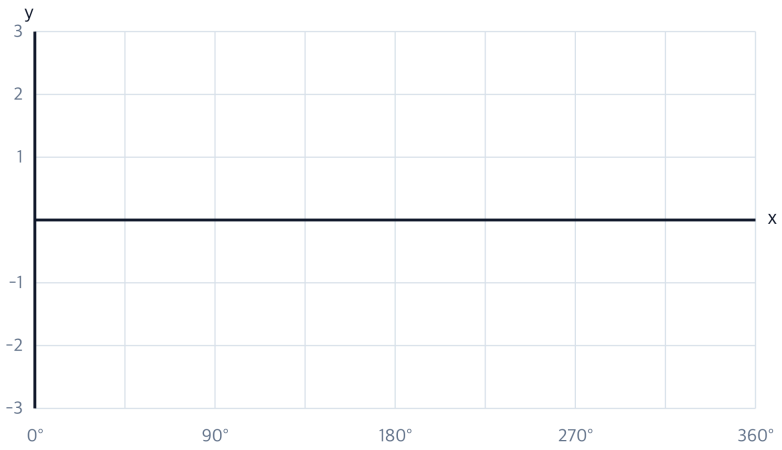
Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1

Sketch each graph and state the period.



Sketch y = cos(2(x)) and state the period.

Answer:

Period = 180°.

Working:

Start from the parent graph y = cos x.

The coefficient outside the trigonometric function is 1, so the amplitude is |1| = 1.

The coefficient inside the bracket is 2, so the period is 360° ÷ 2 = 180°.

The graph is shifted 0° horizontally.

The + 0 outside the function gives the midline y = 0.

Summarise the graph features.

Amplitude	1
Period	180°
Midline	y = 0
Phase shift	none

Use the transformed key points to guide the sketch.

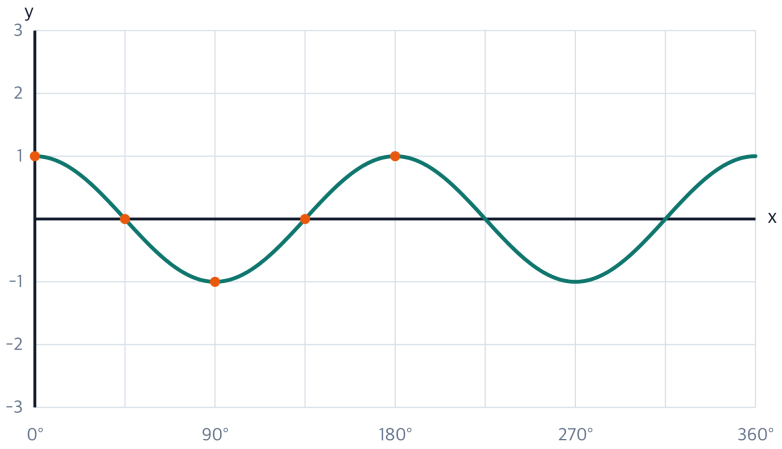
x	y
0°	1
45°	0
90°	-1
135°	0
180°	1

Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1 (continued)



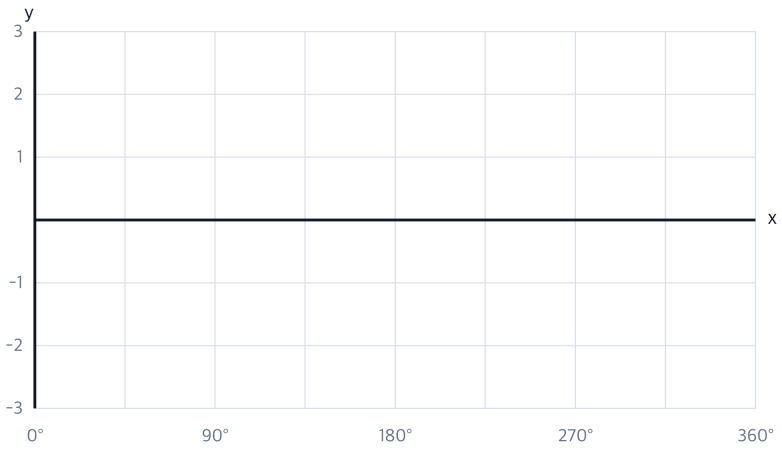
Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1

Sketch each graph and state the midline.



Sketch y = sin(x) - 1 and state the midline.

Answer:

Midline: y = -1.

Working:

Start from the parent graph y = sin x.

The coefficient outside the trigonometric function is 1, so the amplitude is |1| = 1.

The coefficient inside the bracket is 1, so the period is 360° ÷ 1 = 360°.

The graph is shifted 0° horizontally.

The - 1 outside the function gives the midline y = -1.

Summarise the graph features.

Amplitude	1
Period	360°
Midline	y = -1
Phase shift	none

Use the transformed key points to guide the sketch.

x	y
0°	-1
90°	0
180°	-1
270°	-2
360°	-1

Teacher answers & working

Trigonometric Graphs and Transformations · Single Transformations

ALG-020E-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1 (continued)

